

Integration of satellite imagery and cropping systems model to determine major crop yields in data scarce environments

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Abstract: This research introduces an innovative methodology tailored for data-scarce environments to estimate wheat crop grain yield, employing the METRIC-CROPSYST model, satellite imagery, and meteorological data. The study focuses on four distinct Agroecological Zones in Punjab, Pakistan, during the Rabi cropping seasons from 2016 to 2021. Each site exhibited unique characteristics and management practices, including variations in rain-fed and irrigated croplands, as well as differences in seasonal precipitation and average temperatures. The implementation of the METRIC-CROPSYST model provided a satellite-based mechanistic modelling solution, enabling precise biomass and grain yield predictions with minimal in-situ data requirements. Utilizing Landsat 7 and 8 imagery along with METRIC model outputs from the EEFlux application on Google Earth Engine, we processed these outputs for cloud masking, gap-filling, and temporal interpolation, ensuring daily time-step input data accuracy. This approach addressed the common challenges of data gaps and cloud cover in satellite imagery, significantly enhancing the reliability of the data used for crop modelling. Further analyses included crop seasonality determination and actual transpiration estimates, which are critical factors in understanding and predicting crop growth and yield. Comparisons with regional averages for biomass and yield estimation showcased promising results, demonstrating the model's efficacy in capturing growth dynamics accurately. The study highlighted the METRIC-CROPSYST model's overall strong performance, suggesting its potential as an efficient and reliable alternative to traditional methods in regional grain yield estimation. The model's ability to integrate real-time remote sensing data with agro-meteorological information allows for a more robust understanding of crop yields, circumventing the need for extensive site-specific calibration or training datasets typically required by machine learning and statistical models. This research demonstrates the advantages of utilizing advancements in Earth observation technologies such as availability of high-quality and consistent satellite imagery, progress in multi-sensor data fusion techniques and scientific methodologies, combined with a simple crop growth algorithm to accurately estimate crop yield. Moreover, the integration of machine and deep learning techniques for cloud detection and gap-filling in satellite images has further enhanced the quality of the input data, ensuring more reliable crop model outputs. The study's findings have important implications for informed decision-making in agricultural management and resource allocation. By providing dependable yield estimates, the integrated approach facilitates timely and precise assessments of crop yields, which are crucial for shaping food security policies, especially in regions with limited access to ground-based data and monitoring resources. Given the increasing global demand for agricultural products, this methodology presents a viable solution for enhancing food production systems and ensuring food security in data-scarce environments. In conclusion, the METRIC-CROPSYST model's integration with satellite imagery and minimal meteorological data offers a promising strategy for accurate and reliable crop yield estimation. This research paves the way for the model's future adoption in similar contexts, emphasizing its role in enhancing agricultural planning and resource management. The practical insights provided by the study outline key considerations, challenges, and opportunities for the broad application of satellite remote sensing-based cropping systems models across diverse agricultural landscapes.

Keywords: Satellite Imagery, Grain Yield, Biomass, Evapotranspiration, Crop Modelling, Decision Support